Graphs of Sine and Cosine Functions

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Amplitude, period, and midline

For the function $y = 3\sin(2x)$, state:

a the amplitude **b** the period **c** the range

2 [Foundational] • Target: Amplitude, period, and midline

Consider the mathematical condition where for the function $y = -2\cos\left(\frac{x}{2}\right) + 1$, state:

a the amplitude **b** the period

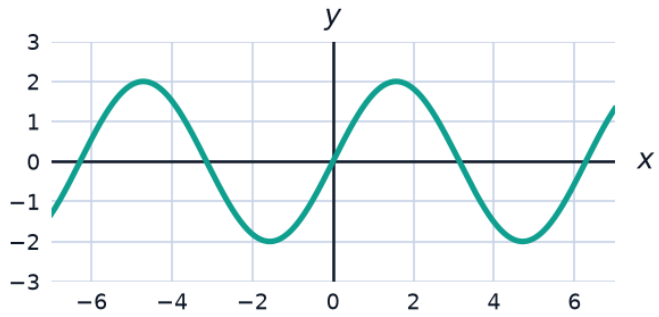
c the midline **d** the range

3 [Foundational] • Target: Amplitude, period, and midline

For $y = \sin\left(x - \frac{\pi}{3}\right)$, state the phase shift and the period.

4 [Foundational] • Target: Reading and writing equations from graphs

The graph below shows one sinusoidal function. State its amplitude and period, then write an equation for the function.



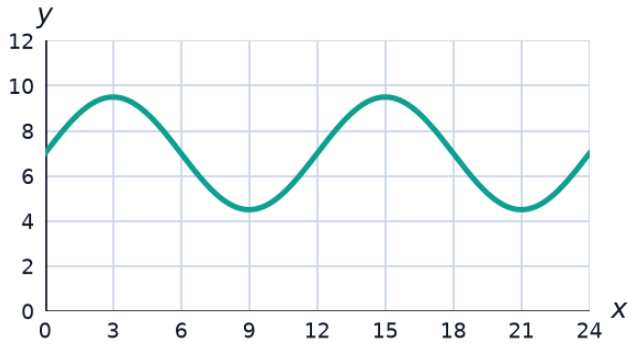
5 [Foundational] • Target: Reading and writing equations from graphs

Write an equation of a cosine function with amplitude 4, period π, reflected across the x-axis, with no phase or vertical shift.

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6 [Foundational] • Target: Reading and writing equations from graphs

The depth of water at a harbour entrance is modeled by $d(t) = 2.5\sin\left(\frac{\pi}{6}t\right) + 7$ metres, where t is in hours after midnight.



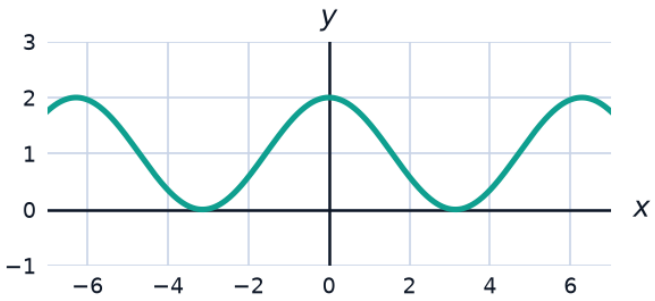
a State the maximum and minimum depths.

b State the period of the tide cycle.

c Find the depth at $t = 3$.

7 [Foundational] • Target: Graphing with transformations

The graph below shows $y = \cos(x) + 1$, a cosine curve with a vertical shift. Identify the midline and the maximum and minimum values shown.

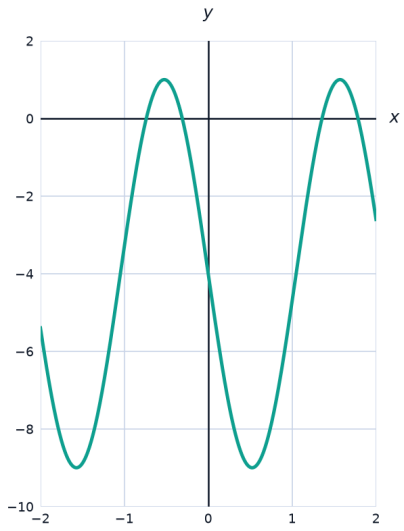


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: Graphing with transformations**

Describe, in order, the transformations that turn $y = \sin x$ into $y = -3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 2$.



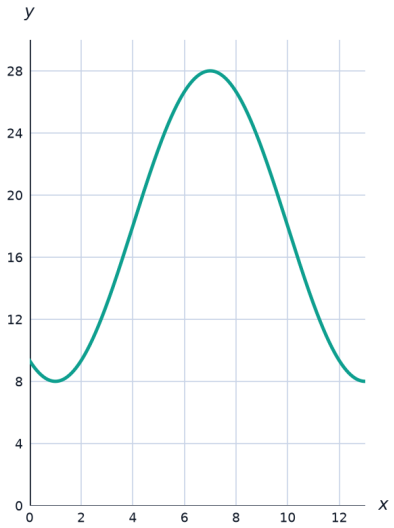
9 **[Intermediate]** • **Target: Graphing with transformations**

State the amplitude, period, phase shift, and midline for $y = 5\cos\left(3x + \frac{\pi}{2}\right) - 4$. (Hint: factor the argument as $3\left(x + \frac{\pi}{6}\right)$.)

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10 [Intermediate] • Target: Modeling periodic phenomena

The average monthly temperature in a city varies between $8°\text{C}$ (in January, month 1) and $28°\text{C}$ (in July, month 7), following a sinusoidal pattern with a 12-month period.



a Find the amplitude and midline of the model.

b Write a cosine model $T(m)$ for the temperature in month m, using the fact that the minimum occurs at $m = 1$.

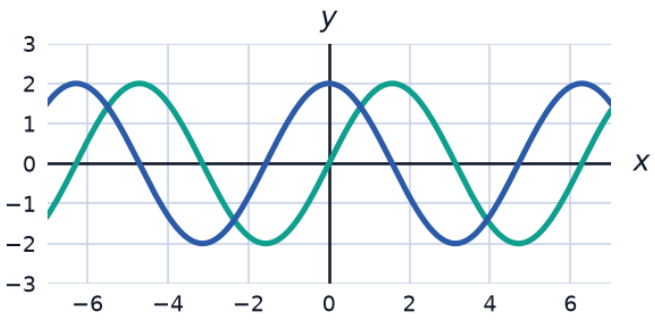
c Use your model to estimate the temperature in month 4 (April).

11 [Intermediate] • Target: Comparing sine and cosine graphs

Explain why $\cos x = \sin\left(x + \frac{\pi}{2}\right)$ by comparing the graphs of sine and cosine.

12 [Intermediate] • Target: Comparing sine and cosine graphs

Both $y = 2\sin(x)$ and $y = 2\cos(x)$ are graphed below. State one key visual difference and one similarity between them.



13 [Intermediate] • Target: Finding zeros and extrema from an equation

Find all zeros of $y = \sin(2x)$ on the interval $[0, 2\pi]$.

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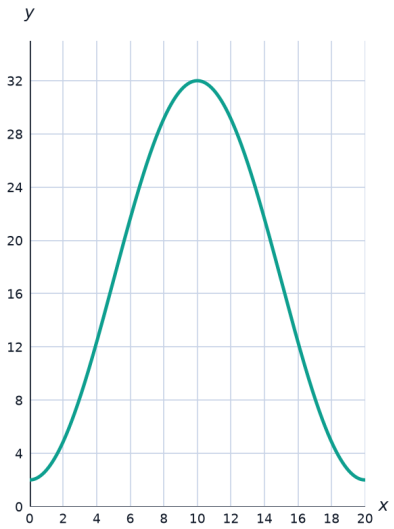
14 [Intermediate] • Target: Finding zeros and extrema from an equation

Find the x-value(s) in $[0, 2\pi]$ where $y = 3\cos(x)$ attains its maximum and minimum values.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

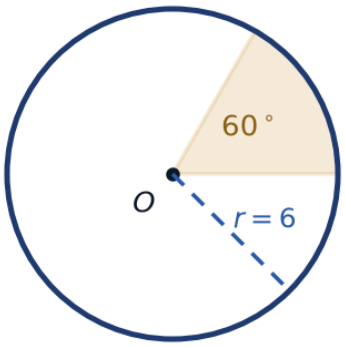
15 [Challenging] • Target: Solving basic sinusoidal equations • Cross-Lesson: 7.02 Logarithms and Their Properties

A Ferris wheel's height above ground is $h(t) = 15\sin\left(\frac{\pi}{10}t - \frac{\pi}{2}\right) + 17$ meters. Find the first time $t > 0$ (minutes) when the rider is at height 17 m and rising.



16 [Challenging] • Target: Reading and writing equations from graphs • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

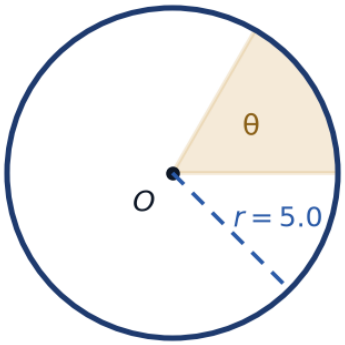
Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.



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17 [Challenging] • Target: Graphing with transformations • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.



18 [Challenging] • Target: Modeling periodic phenomena • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Consider the points $A(1, 2, 3)$, $B(2, 4, 1)$, and $C(-1, 0, 5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.

19 [Challenging] • Target: Comparing sine and cosine graphs • Cross-Lesson: 8.02 Graphing Rational Functions

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

20 [Challenging] • Target: Finding zeros and extrema from an equation • Cross-Lesson: 8.03 Introduction to Limits

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Amplitude, period, and midline

Write a sinusoidal function with amplitude 5, period 4π, midline $y = -3$, using $\cos$.

a. $f(x) = 5\cos(1/2x) - 3$

b. $f(x) = 5\cos(1/2x) - 2$

c. $f(x) = 6\cos(1/2x) - 3$

d. $f(x) = 5\cos(1x) - 3$

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22 [Foundational] • Target: Reading and writing equations from graphs

The function $f(x) = -2\cos(\frac{\pi}{3}x) + 5$ has amplitude, period, and midline

a. 2, 6, y=5

b. 2, 3, y=5

c. –2, 6, y=5

d. 2, 6, y=0

23 [Foundational] • Target: Graphing with transformations

A sinusoidal function has period $\frac{2\pi}{3}$, midline $y = 3$, and minimum value -1. Find its amplitude.

a. 1

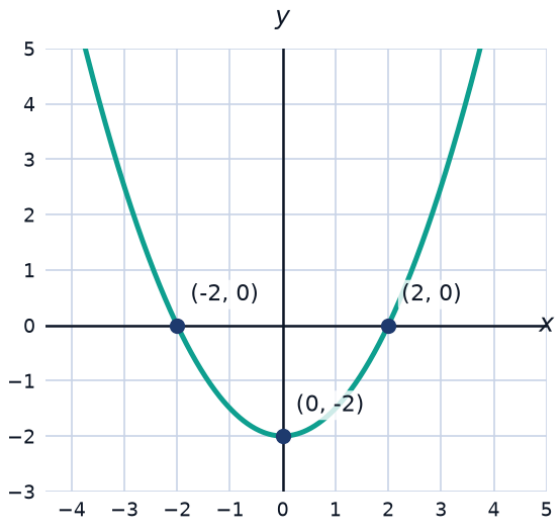
b. 4

c. 9

d. 2

24 [Foundational] • Target: Modeling periodic phenomena

The graph of the quadratic function $f(x) = \frac{1}{2}x^2 - 2$ is shown in the figure below. How many distinct real solutions does the equation $(f \circ f \circ f)(x) = 0$ have?



a. 3

b. 4

c. 6

d. 8

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25 [Foundational] • Target: Comparing sine and cosine graphs

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y-coordinate of the point P?

a. 3

b. 4

c. 5

d. 6

26 [Foundational] • Target: Finding zeros and extrema from an equation

A circle has its center at the origin $(0, 0)$ and passes through the point $P(3, y)$. If the equation of the circle is $x^2 + y^2 = 25$, what is the positive value of y?

a. 3

b. 4

c. 5

d. 16

27 [Foundational] • Target: Solving basic sinusoidal equations

A circle is centered at the origin $(0, 0)$ and passes through the point $(3, 4)$. What is the equation of this circle?

a. $x^2 + y^2 = 7$

b. $x^2 + y^2 = 9$

c. $x^2 + y^2 = 16$

d. $x^2 + y^2 = 25$

28 [Foundational] • Target: Amplitude, period, and midline

What is the radius of the circle with the equation $(x - 3)^2 + (y + 5)^2 = 36$?

a. 6

b. 36

c. 3

d. 5

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Reading and writing equations from graphs

What is the Cartesian equation of the polar curve $r = 1$?

a. $x^2 + y^2 = 1$

b. $x^2 - y^2 = 1$

c. $x^2 + y^2 = 2$

d. x+y=1

30 [Intermediate] • Target: Graphing with transformations

Find the unit vector in the direction of $\mathbf{v} = \langle -8, 15 \rangle$.

a. $\left\langle \frac{9}{17}, \frac{15}{17} \right\rangle$

b. $\left\langle \frac{15}{17}, -\frac{8}{17} \right\rangle$

c. $\left\langle -\frac{8}{17}, \frac{15}{17} \right\rangle$

d. $\langle -8, 15 \rangle$

31 [Intermediate] • Target: Modeling periodic phenomena

A quantity starts at 100 and grows by 10% each period. Find its value after 2 periods.

a. 121

b. 100

c. 126

d. 120

32 [Intermediate] • Target: Comparing sine and cosine graphs

A quantity starts at 100 and decays by 10% each period. Find its value after 2 periods.

a. 81

b. 100

c. 80

d. 86

33 [Intermediate] • Target: Finding zeros and extrema from an equation

A quantity starts at 1600 and is divided by 2 every period. Find its value after 4 periods.

a. 1600

b. 100

c. 102

d. 6400

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34 [Intermediate] • Target: Solving basic sinusoidal equations

A quantity starts at 50 and multiplies by 3 every period. Find its value after 3 periods.

a. 1353

b. 1350

c. 150

d. 50

35 [Intermediate] • Target: Amplitude, period, and midline

A population starts at 100 and doubles every period. Find its value after 3 periods.

a. 800

b. 300

c. 100

d. 802

36 [Intermediate] • Target: Reading and writing equations from graphs

A population starts at 3 and is multiplied by 3 every period. Find its value after 3 periods.

a. 3

b. 84

c. 9

d. 81

37 [Intermediate] • Target: Graphing with transformations

For the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$, what are the equations of the asymptotes?

a. $y = \pm\frac{3}{2}x$

b. $y = \pm\frac{2}{3}x$

c. $y = \pm 6x$

d. $y = \pm\frac{5}{2}x$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Modeling periodic phenomena • Cross-Lesson: 8.01 Rational Functions and Asymptotes

The polar equation $r = 2\cos\theta$ represents a circle. What is its radius?

a. 2

b. 3

c. 5

d. 1

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39 [Challenging] • Target: Comparing sine and cosine graphs • Cross-Lesson: 8.02 Graphing Rational Functions

A matrix has 1 rows and 3 columns. What are its dimensions?

a. 1×3

b. 4

c. 3

d. 3×1

40 [Challenging] • Target: Finding zeros and extrema from an equation • Cross-Lesson: 8.03 Introduction to Limits

Use Cramer's Rule to solve: $5x - 3y = 14,\ -3x + 4y = -4$.

a. $x = 4, y = 3$

b. $x = 4, y = 2$

c. $x = 5, y = 2$

d. $x = 2, y = 4$

41 [Challenging] • Target: Solving basic sinusoidal equations • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Eliminate the parameter: $x = t,\ y = -1t + 5$.

a. $x = -1y + 5$

b. $y = -1x$

c. $y = -1x + 5$

d. $y = 1x + 5$

42 [Challenging] • Target: Amplitude, period, and midline • Cross-Lesson: 9.02 Geometric Sequences and Series

Find the determinant of $(-1, 2;\ -6, 4)$.

a. 8

b. -16

c. 9

d. 7

43 [Challenging] • Target: Reading and writing equations from graphs • Cross-Lesson: 9.03 The Binomial Theorem

What is the sum of all real values of k for which the equation $|x^2 - 6x + 5| = 2x + k$ has exactly three distinct real solutions?

a. -3

b. -1

c. -2

d. -13

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44 [Challenging] • Target: Graphing with transformations • Cross-Lesson: 1.01 Functions and Function Notation

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.

a. c = 3, k = 4

b. c = 5, k = 3

c. c = 3, k = 3

d. c = 5, k = 4

45 [Challenging] • Target: Modeling periodic phenomena • Cross-Lesson: 1.02 Domain and Range

The graph of the sinusoidal function $y = A\sin(Bx + C) + D$ is shown below. The function has a maximum value of 5 and a minimum value of -1. The graph passes through the point $(0, 2)$ and has a period of $\frac{\pi}{2}$. Find the value of A and B. Assume $A > 0$ and $B > 0$.

a. $A = 3, B = 4$

b. $A = 3, B = 2$

c. $A = 6, B = 4$

d. $A = 6, B = 2$

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